Quantitative Aptitude/Technical Abilty — Solutions & Explanations

🎯 Infosys Systems Engineer (SE) | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Total Questions: 10 | **Total Marks:** 10

Section: Technical Ability - Mathematics /Quantitative Aptitude

Problem 1 • An Amount of Rs. 3000 is distributed among A, B and C such that A gets 2/3 rd of what B and C together get and C gets 1/2 of what A and B together get. Find B's sh are.

◯ PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

An Amount of Rs. 3000 is distributed among A, B and C such that A gets 2/3 rd of what B and C together get and C gets 1/2 of what A and B together get. Find B's sh are.

A 650

B 700

C 500

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 2 • **Options:**

◯ PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

A 4

B 5

D None Of these

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 3 • An almirah was sold at a profit of 15%. If its cost had been 5% less and it had been sold for Rs 1,470 less, then the profit would have been 10%. What is the cost price of the almirah?

PRACTICE VARIANT

ID: Q3 Source: Infosys PYQ Level: Infosys SE Standard

An almirah was sold at a profit of 15%. If its cost had been 5% less and it had been sold for Rs 1,470 less, then the profit would have been 10%. What is the cost price of the almirah?

A 14400

C 16100

D 15400

Verified Answer: Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 4 • Seven years ago, the ratio of the ages of A and B was 2: 3. Three years hence, the ratio of their ages will be 5: 7. The sum of their present ages (in years) is:

PRACTICE VARIANT

ID: Q4 Source: Infosys PYQ Level: Infosys SE Standard

Seven years ago, the ratio of the ages of A and B was 2: 3. Three years hence, the ratio of their ages will be 5: 7. The sum of their present ages (in years) is:

A 100

B 200

C 112

Verified Answer: Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 5 • Veer buys 200 guavas for Rs.1,800. Some of these guavas are rotten and are thrown away. He sells the remaining guavas at Rs.14 each and makes a profit of Rs.230. The percentage of guavas thrown away is:

PRACTICE VARIANT

ID: Q5 Source: Infosys PYQ Level: Infosys SE Standard

Veer buys 200 guavas for Rs.1,800. Some of these guavas are rotten and are thrown away. He sells the remaining guavas at Rs.14 each and makes a profit of Rs.230. The percentage of guavas thrown away is:

A 27%

B 29%

C 30%

Verified Answer: Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 6 • A boy walked along two adjacent sides of a rectangular field. If he had walked along the diagonal, then he would have saved a distance equal to one-fourth of the larger side. The ratio of the smaller to the larger side is:

◯ PRACTICE VARIANT

ID: Q6 Source: Infosys PYQ Level: Infosys SE Standard

A boy walked along two adjacent sides of a rectangular field. If he had walked along the diagonal, then he would have saved a distance equal to one-fourth of the larger side. The ratio of the smaller to the larger side is:

A 11:18

B 18:11

D 5:12

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 7 • **Options:**

◯ PRACTICE VARIANT

ID: Q7 Source: Infosys PYQ Level: Infosys SE Standard

A 5

B 0

D 4

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 8 • A team of four players are to be selected from among four girls A,B,C,D and four boys P,Q,R,S such that there would be at least two girls in the team and the team selection should meet the following conditions i. B cannot go with R ii .D and P must go together iii S cannot be put up with A iv. R and C must be together

◯ PRACTICE VARIANT

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

A team of four players are to be selected from among four girls A,B,C,D and four boys P,Q,R,S such that there would be at least two girls in the team and the team selection should meet the following conditions i. B cannot go with R ii .D and P must go together iii S cannot be put up with A iv. R and C must be together

A QBSC

C RCQS

D BPRA

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 9 • While Balbir had his back turned, a dog ran into his butcher shop, snatched a piece of meat off the counter and ran off. Balbir was mad when he realised what had happened. He asked three other shopkeepers, who had seen the dog, to describe it. The shopkeepers really didn't want to help Balbir. So each of them made a statement which contained one truth and one lie. A. Shopkeeper Number 1 said: "The dog had black hair and a long tail." B. Shopkeeper Number 2 said: "The dog had a short tail and wore a collar." C. Shopkeeper Number 3 said: "The dog had white hair and no collar." Based on the above statements, which of the following could be a correct description?

◯ PRACTICE VARIANT

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

While Balbir had his back turned, a dog ran into his butcher shop, snatched a piece of meat off the counter and ran off. Balbir was mad when he realised what had happened. He asked three other shopkeepers, who had seen the dog, to describe it. The shopkeepers really didn't want to help Balbir. So each of them made a statement which contained one truth and one lie. A. Shopkeeper Number 1 said: "The dog had black hair and a long tail." B. Shopkeeper Number 2 said: "The dog had a short tail and wore a collar." C. Shopkeeper Number 3 said: "The dog had white hair and no collar." Based on the above statements, which of the following could be a correct description?

A The dog had black hair, long tail and a collar.

B The dog had white hair, short tail and no collar.

C The dog had black hair, long tail and no collar.

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 10 • In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there?

◯ PRACTICE VARIANT

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there?

A 50

B 100

D 200

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.